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OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
90  def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

powershell

Code

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"

CN A03 - Reliable Data Transfer Protocols --- rdt 3.0  |  Go-Back-N  |  Selective Repeat


  Config:
    Packets      : 10
    Packet size  : 8 bytes
    Window size  : 4
    Timeout      : 2.0s
    Loss prob    : 15%
    Corrupt prob : 8%
    Delay prob   : 10%

Choose protocol:
1. rdt 3.0  (Stop-and-Wait)
2. Go-Back-N
3. Selective Repeat
4. Run all three

Enter choice [1-4]: 1

Choose test scenario:
  1. No errors (ideal channel)
  2. Packet loss only
  3. Corruption only
  4. Delayed packets only
  5. All errors (loss + corruption + delay)

Enter scenario [1-5]: 2

[TEST] Running: Packet loss only

----------------------------------------------------
rdt 3.0  SENDER  started  (Stop-and-Wait)
----------------------------------------------------

[SENDER] Sending seq=0  data=b'PKT0000_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0000_'
[CHANNEL] X ACK seq=0 LOST
[SENDER] - TIMEOUT -- retransmitting seq=0

[SENDER] Sending seq=0  data=b'PKT0000_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0000_'
[SENDER] ✓ ACK 0 received -- moving on

[SENDER] Sending seq=1  data=b'PKT0001_'
[CHANNEL] X DATA seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0001_'
```

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OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
90  def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"
[TEST] Running: Packet loss only

------------------------------------------------------
rdt 3.0  SENDER  started  (Stop-and-Wait)
------------------------------------------------------

[SENDER] Sending seq=0  data=b'PKT0000_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0000_'
[CHANNEL] X ACK seq=0 LOST
[SENDER] - TIMEOUT -- retransmitting seq=0

[SENDER] Sending seq=0  data=b'PKT0000_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0000_'
[SENDER] ✓ ACK 0 received -- moving on

[SENDER] Sending seq=1  data=b'PKT0001_'
[CHANNEL] X DATA seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0001_'
[CHANNEL] X DATA seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0001_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0001_'
[CHANNEL] X ACK seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0001_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0001_'
[SENDER] ✓ ACK 1 received -- moving on

[SENDER] Sending seq=0  data=b'PKT0002_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0002_'
[CHANNEL] X ACK seq=0 LOST
[SENDER] - TIMEOUT -- retransmitting seq=0

[SENDER] Sending seq=0  data=b'PKT0002_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0002_'
[SENDER] ✓ ACK 0 received -- moving on

[SENDER] Sending seq=1  data=b'PKT0003_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0003_'
[CHANNEL] X ACK seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0003_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0003_'
[SENDER] ✓ ACK 1 received -- moving on
```

powershell

Code

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rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"
[SENDER] ✓ ACK 0 received -- moving on

[SENDER] Sending seq=1  data=b'PKT0005_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0005_'
[CHANNEL] X ACK seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0005_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0005_'
[SENDER] ✓ ACK 1 received -- moving on

[SENDER] Sending seq=0  data=b'PKT0006_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0006_'
[SENDER] ✓ ACK 0 received -- moving on

[SENDER] Sending seq=1  data=b'PKT0007_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0007_'
[SENDER] ✓ ACK 1 received -- moving on

[SENDER] Sending seq=0  data=b'PKT0008_'
[CHANNEL] X DATA seq=0 LOST
[SENDER] - TIMEOUT -- retransmitting seq=0

[SENDER] Sending seq=0  data=b'PKT0008_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0008_'
[SENDER] ✓ ACK 0 received -- moving on

[SENDER] Sending seq=1  data=b'PKT0009_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0009_'
[CHANNEL] X ACK seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0009_'
[CHANNEL] X DATA seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0009_'
[CHANNEL] X DATA seq=1 LOST
[SENDER] - TIMEOUT -- retransmitting seq=1

[SENDER] Sending seq=1  data=b'PKT0009_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0009_'
[SENDER] ✓ ACK 1 received -- moving on

[SENDER] All 10 packets delivered.
[SENDER] Total retransmissions: 11

Channel stats -> sent=37, lost=11, corrupted=0, delayed=0

Total simulation time: 10.02s
```

powershell

Code

OUTLINE

TIMELINE

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OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
90  def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"

CN A03 - Reliable Data Transfer Protocols --- rdt 3.0  |  Go-Back-N  |  Selective Repeat


  Config:
    Packets      : 10
    Packet size  : 8 bytes
    Window size  : 4
    Timeout      : 2.0s
    Loss prob    : 15%
    Corrupt prob : 8%
    Delay prob   : 10%

Choose protocol:
1. rdt 3.0  (Stop-and-Wait)
2. Go-Back-N
3. Selective Repeat
4. Run all three

Enter choice [1-4]: 2

Choose test scenario:
  1. No errors (ideal channel)
  2. Packet loss only
  3. Corruption only
  4. Delayed packets only
  5. All errors (loss + corruption + delay)

Enter scenario [1-5]: 3

[TEST] Running: Corruption only

----------------------------------------------------
Go-Back-N  SENDER  (N=4)
----------------------------------------------------

[SENDER] Sending seq=0  (window: 0-3)
[CHANNEL] X DATA seq=0 CORRUPTED
[RECEIVER] X Corrupt packet seq=0 -- discarded

[SENDER] Sending seq=1  (window: 0-3)
[RECEIVER] X Out-of-order seq=1 (expected 0) -- discarded

[SENDER] Sending seq=2  (window: 0-3)
[RECEIVER] X Out-of-order seq=2 (expected 0) -- discarded

[SENDER] Sending seq=3  (window: 0-3)
[CHANNEL] X DATA seq=3 CORRUPTED
[RECEIVER] X Corrupt packet seq=3 -- discarded
```

powershell

Code

0 0

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rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
90  def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"

[SENDER] TIMEOUT -- Go-Back-N: retransmitting 5..8
[RECEIVER] ✓ In-order seq=5 accepted
[RECEIVER] ✓ In-order seq=6 accepted
[CHANNEL] X DATA seq=7 CORRUPTED
[RECEIVER] X Corrupt packet seq=7 -- discarded
[CHANNEL] X DATA seq=8 CORRUPTED
[RECEIVER] X Corrupt packet seq=8 -- discarded
[SENDER] ✓ Cumulative ACK 5 received

[SENDER] Sending seq=9  (window: 6-9)
[RECEIVER] X Out-of-order seq=9 (expected 7) -- discarded
[SENDER] ✓ Cumulative ACK 6 received
[SENDER] (duplicate/old ACK 6 -- ignored)
[SENDER] (duplicate/old ACK 6 -- ignored)
[SENDER] (duplicate/old ACK 6 -- ignored)

[SENDER] TIMEOUT -- Go-Back-N: retransmitting 7..9
[CHANNEL] X DATA seq=7 CORRUPTED
[RECEIVER] X Corrupt packet seq=7 -- discarded
[RECEIVER] X Out-of-order seq=8 (expected 7) -- discarded
[CHANNEL] X DATA seq=9 CORRUPTED
[RECEIVER] X Corrupt packet seq=9 -- discarded
[SENDER] (duplicate/old ACK 6 -- ignored)
[SENDER] (duplicate/old ACK 6 -- ignored)
[SENDER] (duplicate/old ACK 6 -- ignored)

[SENDER] TIMEOUT -- Go-Back-N: retransmitting 7..9
[RECEIVER] ✓ In-order seq=7 accepted
[CHANNEL] X ACK seq=7 CORRUPTED
[CHANNEL] X DATA seq=8 CORRUPTED
[RECEIVER] X Corrupt packet seq=8 -- discarded
[RECEIVER] X Out-of-order seq=9 (expected 8) -- discarded
[SENDER] X Corrupt ACK -- ignored
[SENDER] ✓ Cumulative ACK 7 received
[SENDER] (duplicate/old ACK 7 -- ignored)

[SENDER] TIMEOUT -- Go-Back-N: retransmitting 8..9
[RECEIVER] ✓ In-order seq=8 accepted
[RECEIVER] ✓ In-order seq=9 accepted
[SENDER] ✓ Cumulative ACK 8 received
[SENDER] ✓ Cumulative ACK 9 received

[SENDER] All 10 packets delivered.
[SENDER] Total retransmissions: 28

Channel stats -> sent=72, lost=0, corrupted=23, delayed=0

Total simulation time: 16.78s
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments>
```

powershell

Code

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OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
90  def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

powershell

Code

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"

CN A03 - Reliable Data Transfer Protocols --- rdt 3.0  |  Go-Back-N  |  Selective Repeat


  Config:
    Packets      : 10
    Packet size  : 8 bytes
    Window size  : 4
    Timeout      : 2.0s
    Loss prob    : 15%
    Corrupt prob : 8%
    Delay prob   : 10%

Choose protocol:
1. rdt 3.0  (Stop-and-Wait)
2. Go-Back-N
3. Selective Repeat
4. Run all three

Enter choice [1-4]: 3

Choose test scenario:
  1. No errors (ideal channel)
  2. Packet loss only
  3. Corruption only
  4. Delayed packets only
  5. All errors (loss + corruption + delay)

Enter scenario [1-5]: 4

[TEST] Running: Delayed packets only

--------------------------------------------------
Selective Repeat  SENDER  (N=4)
--------------------------------------------------

[SENDER] Sending seq=0  (window: 0-3)
[RECEIVER] ✓ Buffered seq=0
[RECEIVER] ^ Delivered seq=0  data=b'PKT0000_'  to upper layer
[CHANNEL] ACK seq=0 DELAYED 0.03s

[SENDER] Sending seq=1  (window: 0-3)
[CHANNEL] DATA seq=1 DELAYED 0.03s
[RECEIVER] ✓ Buffered seq=1
[RECEIVER] ^ Delivered seq=1  data=b'PKT0001_'  to upper layer

[SENDER] Sending seq=2  (window: 0-3)
[RECEIVER] ✓ Buffered seq=2
[RECEIVER] ^ Delivered seq=2  data=b'PKT0002_'  to upper layer
[CHANNEL] ACK seq=2 DELAYED 0.29s
```

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OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"
[RECEIVER] ✓ Buffered seq=3
[RECEIVER] ^ Delivered seq=3  data=b'PKT0003_'  to upper layer
[SENDER] ✓ Individual ACK 0

[SENDER] Sending seq=4  (window: 1-4)
[RECEIVER] ✓ Buffered seq=4
[RECEIVER] ^ Delivered seq=4  data=b'PKT0004_'  to upper layer
[CHANNEL] ACK seq=4 DELAYED 0.03s
[SENDER] ✓ Individual ACK 1

[SENDER] Sending seq=5  (window: 2-5)
[CHANNEL] DATA seq=5 DELAYED 0.10s
[RECEIVER] ✓ Buffered seq=5
[RECEIVER] ^ Delivered seq=5  data=b'PKT0005_'  to upper layer
[CHANNEL] ACK seq=5 DELAYED 0.24s
[SENDER] ✓ Individual ACK 2

[SENDER] Sending seq=6  (window: 3-6)
[RECEIVER] ✓ Buffered seq=6
[RECEIVER] ^ Delivered seq=6  data=b'PKT0006_'  to upper layer
[SENDER] ✓ Individual ACK 3

[SENDER] Sending seq=7  (window: 4-7)
[CHANNEL] DATA seq=7 DELAYED 0.13s
[RECEIVER] ✓ Buffered seq=7
[RECEIVER] ^ Delivered seq=7  data=b'PKT0007_'  to upper layer
[CHANNEL] ACK seq=7 DELAYED 0.02s
[SENDER] ✓ Individual ACK 4

[SENDER] Sending seq=8  (window: 5-8)
[RECEIVER] ✓ Buffered seq=8
[RECEIVER] ^ Delivered seq=8  data=b'PKT0008_'  to upper layer
[SENDER] ✓ Individual ACK 5

[SENDER] Sending seq=9  (window: 6-9)
[CHANNEL] DATA seq=9 DELAYED 0.25s
[RECEIVER] ✓ Buffered seq=9
[RECEIVER] ^ Delivered seq=9  data=b'PKT0009_'  to upper layer
[SENDER] ✓ Individual ACK 6
[SENDER] ✓ Individual ACK 7
[SENDER] ✓ Individual ACK 8
[SENDER] ✓ Individual ACK 9

[SENDER] All 10 packets delivered.
[SENDER] Total retransmissions: 0

Channel stats -> sent=20, lost=0, corrupted=0, delayed=10

Total simulation time: 1.70s
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments>
```

powershell

Code

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OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
90  def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

powershell

Code

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"

CN A03 - Reliable Data Transfer Protocols --- rdt 3.0  |  Go-Back-N  |  Selective Repeat


  Config:
    Packets      : 10
    Packet size  : 8 bytes
    Window size  : 4
    Timeout      : 2.0s
    Loss prob    : 15%
    Corrupt prob : 8%
    Delay prob   : 10%

Choose protocol:
1. rdt 3.0  (Stop-and-Wait)
2. Go-Back-N
3. Selective Repeat
4. Run all three

Enter choice [1-4]: 4

Choose test scenario:
  1. No errors (ideal channel)
  2. Packet loss only
  3. Corruption only
  4. Delayed packets only
  5. All errors (loss + corruption + delay)

Enter scenario [1-5]: 5

[TEST] Running: All errors (loss + corruption + delay)

----------------------------------------------------
rdt 3.0  SENDER  started  (Stop-and-Wait)
----------------------------------------------------

[SENDER] Sending seq=0  data=b'PKT0000_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0000_'
[CHANNEL] X ACK seq=0 LOST
[SENDER] - TIMEOUT -- retransmitting seq=0

[SENDER] Sending seq=0  data=b'PKT0000_'
[CHANNEL] DATA seq=0 DELAYED 0.16s
[RECEIVER] ✓ Received seq=0  data=b'PKT0000_'
[CHANNEL] X ACK seq=0 LOST
[SENDER] - TIMEOUT -- retransmitting seq=0

[SENDER] Sending seq=0  data=b'PKT0000_'
[CHANNEL] X DATA seq=0 CORRUPTED
[CHANNEL] DATA seq=0 DELAYED 0.13s
```

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rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"
[SENDER] Sending seq=0  data=b'PKT0008_'
[RECEIVER] ✓ Received seq=0  data=b'PKT0008_'
[SENDER] ✓ ACK 0 received -- moving on

[SENDER] Sending seq=1  data=b'PKT0009_'
[RECEIVER] ✓ Received seq=1  data=b'PKT0009_'
[CHANNEL] ACK seq=1 DELAYED 0.13s
[SENDER] ✓ ACK 1 received -- moving on

[SENDER] All 10 packets delivered.
[SENDER] Total retransmissions: 10

Channel stats -> sent=38, lost=6, corrupted=4, delayed=6

--------------------------------------------------

--------------------------------------------------
Go-Back-N  SENDER  (N=4)
--------------------------------------------------

[SENDER] Sending seq=0  (window: 0-3)
[RECEIVER] ✓ In-order seq=0 accepted

[SENDER] Sending seq=1  (window: 0-3)
[CHANNEL] X DATA seq=1 CORRUPTED
[RECEIVER] X Corrupt packet seq=1 -- discarded

[SENDER] Sending seq=2  (window: 0-3)
[RECEIVER] X Out-of-order seq=2 (expected 1) -- discarded
[CHANNEL] X ACK seq=0 CORRUPTED
[CHANNEL] ACK seq=0 DELAYED 0.07s

[SENDER] Sending seq=3  (window: 0-3)
[CHANNEL] DATA seq=3 DELAYED 0.01s
[RECEIVER] X Out-of-order seq=3 (expected 1) -- discarded
[SENDER] ✓ Cumulative ACK 0 received

[SENDER] Sending seq=4  (window: 1-4)
[CHANNEL] X DATA seq=4 LOST
[SENDER] (duplicate/old ACK 0 -- ignored)
[SENDER] X Corrupt ACK -- ignored
[SENDER] (duplicate/old ACK 0 -- ignored)

[SENDER] TIMEOUT -- Go-Back-N: retransmitting 1..4
[RECEIVER] ✓ In-order seq=1 accepted
[CHANNEL] X DATA seq=2 LOST
[RECEIVER] X Out-of-order seq=3 (expected 2) -- discarded
[RECEIVER] X Out-of-order seq=4 (expected 2) -- discarded
[CHANNEL] X ACK seq=1 CORRUPTED
[SENDER] ✓ Cumulative ACK 1 received
```

powershell

Code

OUTLINE

TIMELINE

0 0 | Ln 133, Col 11 | Spaces: 4 | UTF-8 | CRLF | Python | Inline suggestions quota reached | 3.13.5 | Python 3.13.5 | Go Live | Prettier

File Edit Selection View Go Run Terminal Help

Assignments

Update

EXPLORER

ASSIGNMENTS

- 01
- 02
- 03
  - __pycache__
  - Assignment-III.pdf
  - config.py
  - gbn.py
  - main.py
  - network.py
  - rdt-protocols.py
  - rdt30.py
  - sr.py
- Adobe Premiere Pro A...
- 2026-03-04 22-14-33....
- Untitled.prproj

OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"
[SENDER] TIMEOUT -- Go-Back-N: retransmitting 6..9
[RECEIVER] ✓ In-order seq=6 accepted
[RECEIVER] ✓ In-order seq=7 accepted
[RECEIVER] ✓ In-order seq=8 accepted
[RECEIVER] ✓ In-order seq=9 accepted
[CHANNEL] X ACK seq=9 LOST
[SENDER] ✓ Cumulative ACK 6 received
[SENDER] ✓ Cumulative ACK 7 received
[SENDER] ✓ Cumulative ACK 8 received

[SENDER] TIMEOUT -- Go-Back-N: retransmitting 9..9
[RECEIVER] X Out-of-order seq=9 (expected 10) -- discarded
[SENDER] ✓ Cumulative ACK 9 received

[SENDER] All 10 packets delivered.
[SENDER] Total retransmissions: 13

Channel stats -> sent=43, lost=6, corrupted=3, delayed=2

--------------------------------------------------


--------------------------------------------------
Selective Repeat  SENDER  (N=4)
--------------------------------------------------

[SENDER] Sending seq=0  (window: 0-3)
[CHANNEL] X DATA seq=0 LOST

[SENDER] Sending seq=1  (window: 0-3)
[RECEIVER] ✓ Buffered seq=1

[SENDER] Sending seq=2  (window: 0-3)
[RECEIVER] ✓ Buffered seq=2

[SENDER] Sending seq=3  (window: 0-3)
[RECEIVER] ✓ Buffered seq=3
[SENDER] ✓ Individual ACK 1
[SENDER] ✓ Individual ACK 2
[SENDER] ✓ Individual ACK 3

[SENDER] TIMEOUT for seq=0 -- SR: retransmitting ONLY seq=0
[RECEIVER] ✓ Buffered seq=0
[RECEIVER] ^ Delivered seq=0  data=b'PKT0000_'  to upper layer
[RECEIVER] ^ Delivered seq=1  data=b'PKT0001_'  to upper layer
[RECEIVER] ^ Delivered seq=2  data=b'PKT0002_'  to upper layer
[RECEIVER] ^ Delivered seq=3  data=b'PKT0003_'  to upper layer
[SENDER] ✓ Individual ACK 0

[SENDER] Sending seq=4  (window: 4-7)
[CHANNEL] X DATA seq=4 CORRUPTED
```

powershell

Code

0 0 | Ln 133, Col 11 | Spaces: 4 | UTF-8 | CRLF | Python | Inline suggestions quota reached | 3.13.5 | Python 3.13.5 | Go Live | Prettier

File Edit Selection View Go Run Terminal Help

Assignments

Update

EXPLORER

ASSIGNMENTS

- 01
- 02
- 03
  - __pycache__
  - Assignment-III.pdf
  - config.py
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  - sr.py
- Adobe Premiere Pro A...
- 2026-03-04 22-14-33....
- Untitled.prproj

OUTLINE

TIMELINE

rdt-protocols.py | config.py | network.py | rdt30.py | gbn.py | sr.py | main.py

03 > main.py > ...

```
def run_protocol(choice: str, channel: UnreliableChannel):
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL

```
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments> python -u "c:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments\03\main.py"
[RECEIVER] ✓ Buffered seq=6

[SENDER] Sending seq=7  (window: 4-7)
[RECEIVER] ✓ Buffered seq=7
[SENDER] ✓ Individual ACK 6
[SENDER] ✓ Individual ACK 7

[SENDER] TIMEOUT for seq=4 -- SR: retransmitting ONLY seq=4
[RECEIVER] ✓ Buffered seq=4
[RECEIVER] ^ Delivered seq=4  data=b'PKT0004_'  to upper layer

[SENDER] TIMEOUT for seq=5 -- SR: retransmitting ONLY seq=5
[RECEIVER] ✓ Buffered seq=5
[RECEIVER] ^ Delivered seq=5  data=b'PKT0005_'  to upper layer
[RECEIVER] ^ Delivered seq=6  data=b'PKT0006_'  to upper layer
[RECEIVER] ^ Delivered seq=7  data=b'PKT0007_'  to upper layer
[SENDER] ✓ Individual ACK 4

[SENDER] Sending seq=8  (window: 5-8)
[RECEIVER] ✓ Buffered seq=8
[RECEIVER] ^ Delivered seq=8  data=b'PKT0008_'  to upper layer
[SENDER] ✓ Individual ACK 5

[SENDER] Sending seq=9  (window: 8-9)
[RECEIVER] ✓ Buffered seq=9
[RECEIVER] ^ Delivered seq=9  data=b'PKT0009_'  to upper layer
[CHANNEL] X ACK seq=9 LOST
[SENDER] ✓ Individual ACK 8

[SENDER] TIMEOUT for seq=9 -- SR: retransmitting ONLY seq=9
[CHANNEL] X DATA seq=9 LOST

[SENDER] TIMEOUT for seq=9 -- SR: retransmitting ONLY seq=9
[CHANNEL] X DATA seq=9 CORRUPTED
[RECEIVER] X Corrupt packet seq=9 -- discarded (no ACK)

[SENDER] TIMEOUT for seq=9 -- SR: retransmitting ONLY seq=9
[CHANNEL] X DATA seq=9 LOST

[SENDER] TIMEOUT for seq=9 -- SR: retransmitting ONLY seq=9
[RECEIVER] (old packet seq=9 -- re-ACKing)
[SENDER] ✓ Individual ACK 9

[SENDER] All 10 packets delivered.
[SENDER] Total retransmissions: 7

Channel stats -> sent=28, lost=4, corrupted=3, delayed=0

Total simulation time: 26.16s
PS C:\Ibrahim\Personal\University Stuff\Computer Networks\Assignments>
```

powershell

Code

0 0 | Ln 133, Col 11 | Spaces: 4 | UTF-8 | CRLF | Python | Inline suggestions quota reached | 3.13.5 | Python 3.13.5 | Go Live | Prettier